

Time Series Analysis

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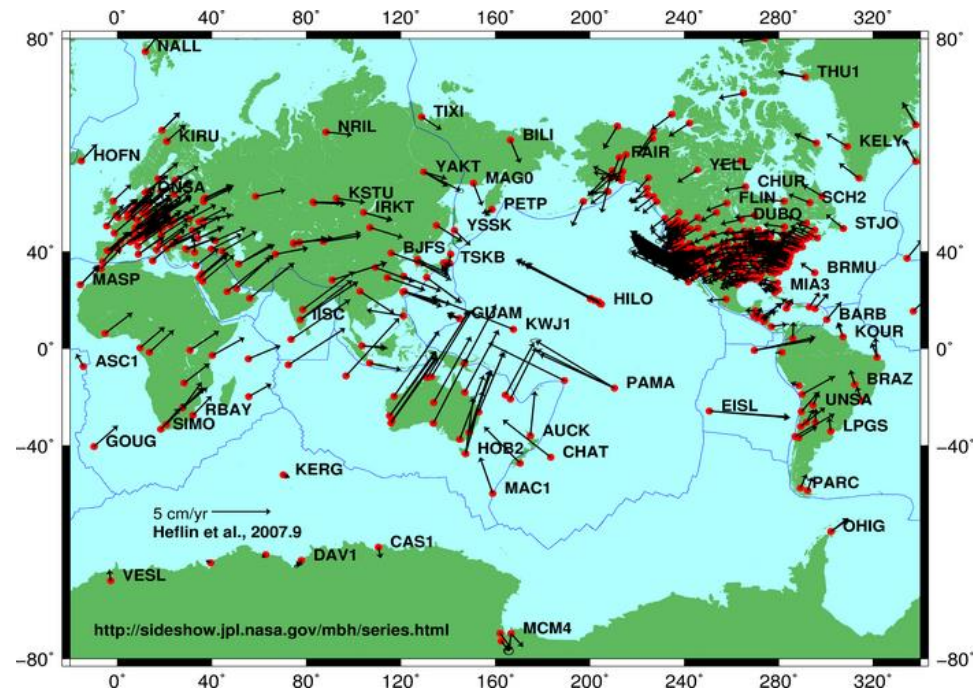
- Introduction to time series analysis
- Examples of different time series
- Stationary and Ergodicity
- Autocovariance and autocorrelation
- Different noise process



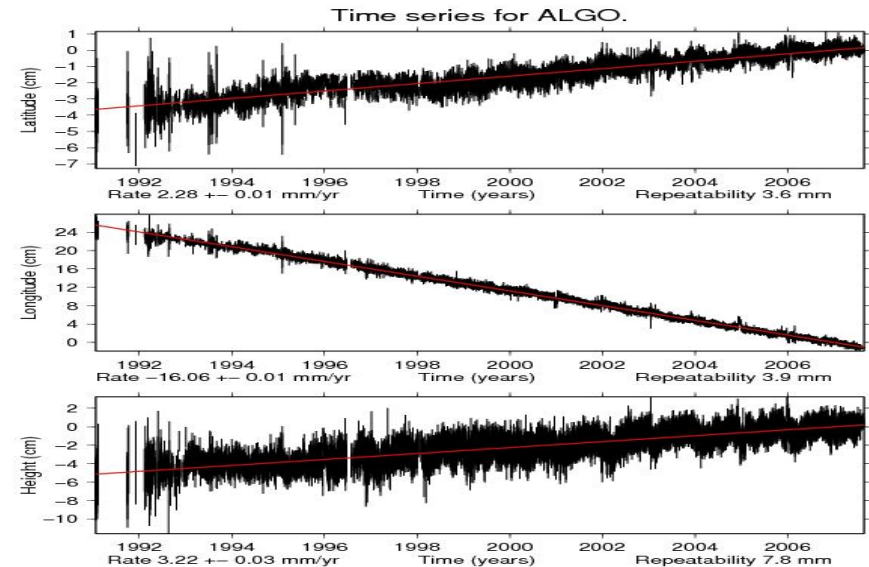
Examples of Time Series

GPS position time series:

Global velocities: IGS stations



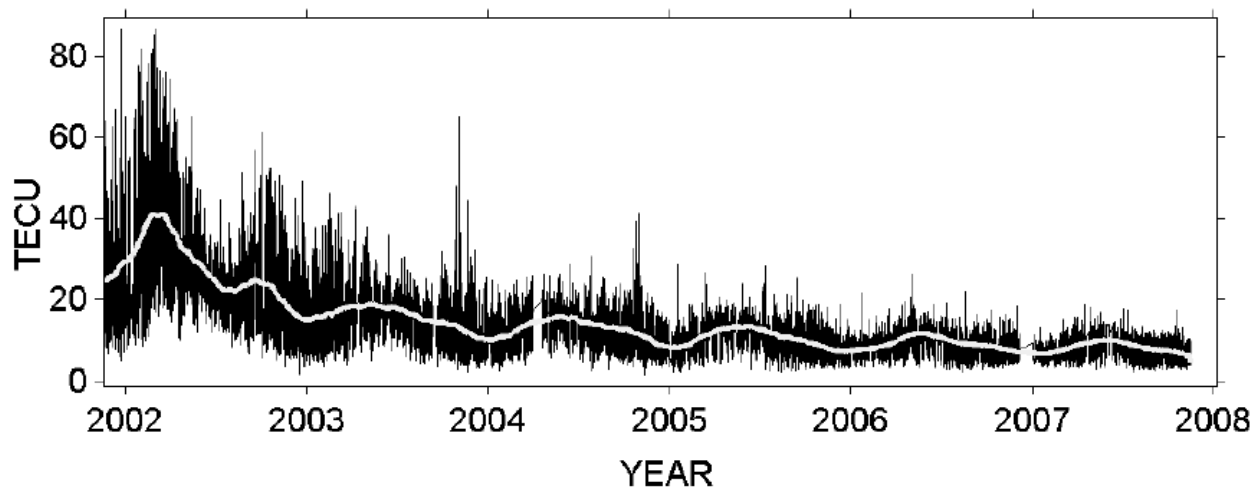
ALGO Station in Canada



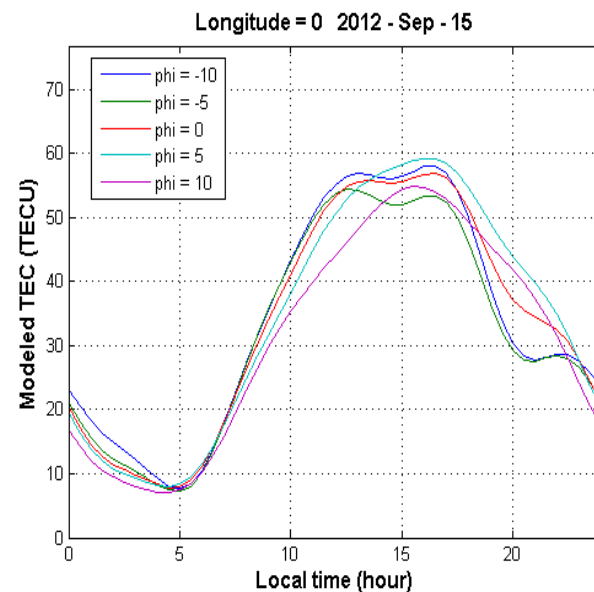
Examples of Time Series

TEC time series:

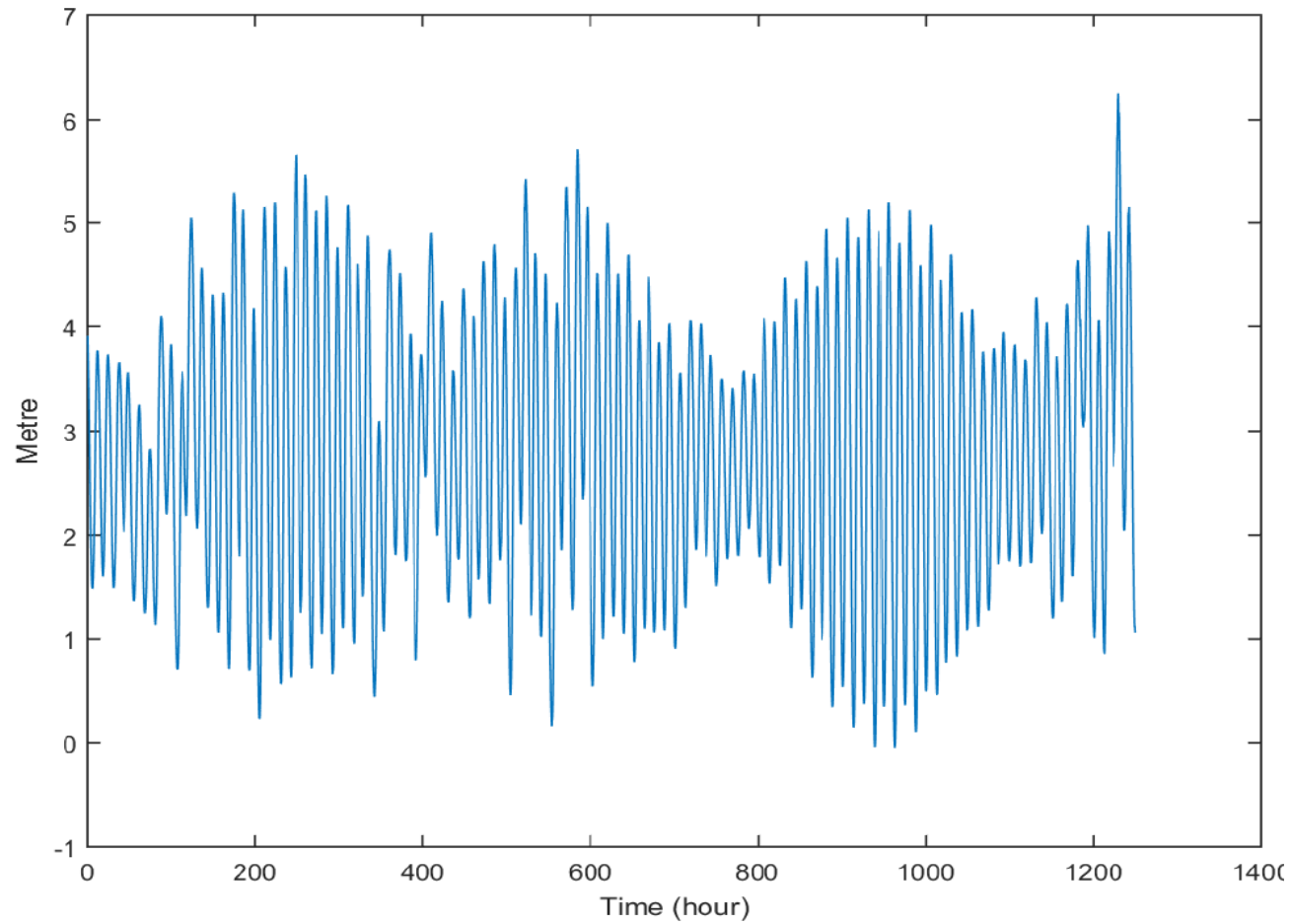
Long term variations



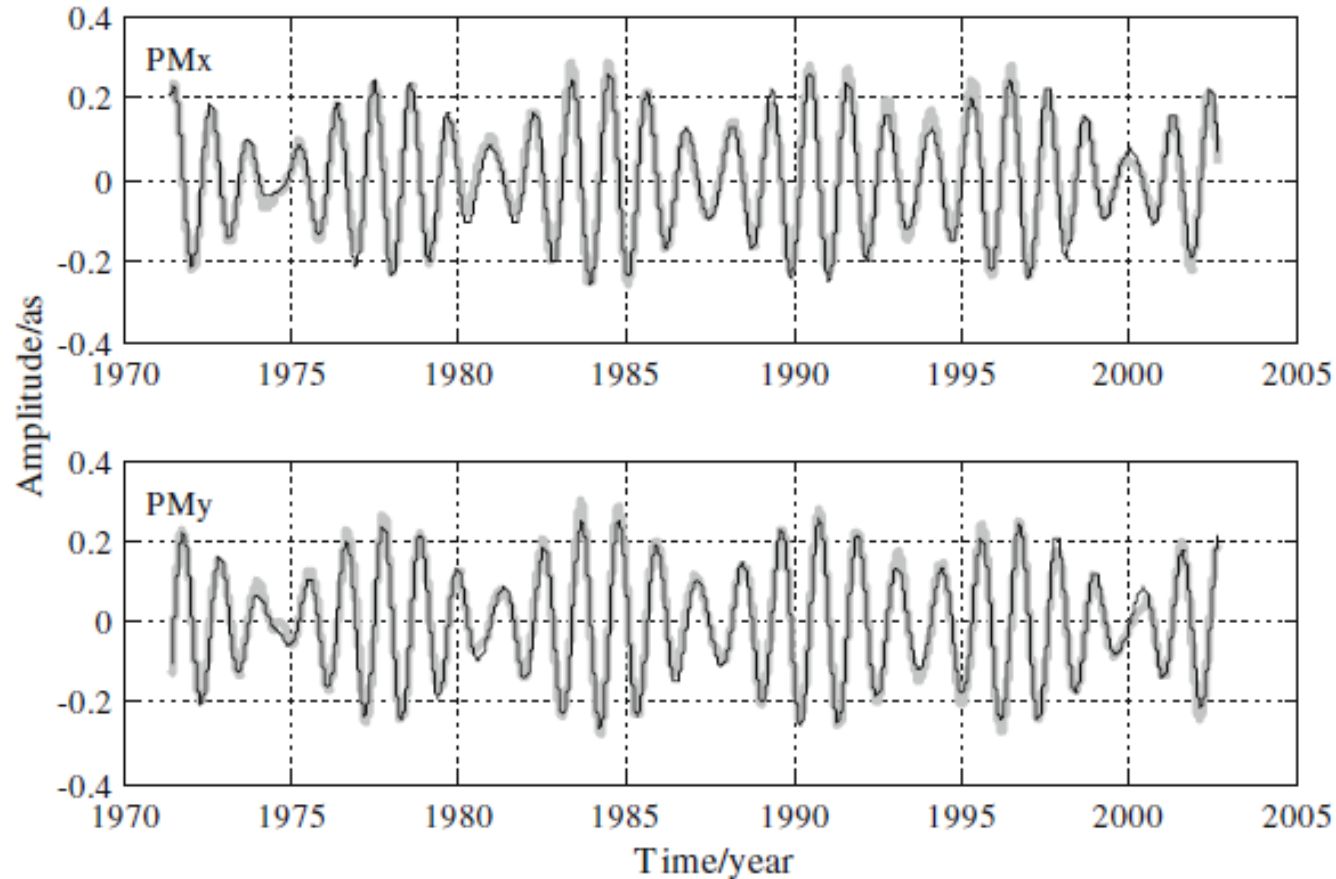
Daily variation of TEC



Sea level time series



Earth orientation parameters time series



Types of time series

Continuous vs. discrete

- Observations made continually in time give rise to a *Continuous Time Series*
- More often, observations are taken only at specific points in time, giving rise to a *Discrete Time Series*



Types of time series

Univariate vs. multivariate

- An analysis of a single sequence of data is called univariate time-series analysis
- An analysis of several sets of data for the same sequence of time periods is called multivariate time-series analysis or, more simply, multiple time-series analysis



Basic time series

Data on the outcome of a variable or variables in different time periods are known as time-series data. A set of observations indexed by time t Discrete and continuous time series

- Time-series data are usually challenging because they are likely to violate the underlying assumptions of linear regression
- Residual errors are correlated instead of being uncorrelated, leading to inconsistent coefficient estimates
- The mean and/or variance of the explanatory variables may change over time, leading to invalid regression results.
- Example of a basic time series known as an autoregressive process



Components of time series

We may in general distinguish the following in a time series

- A trend → general behavior and variation of the process
- Periodical phenomena → e.g. pure seasonal or daily variations
- Irregularities → Generally due to unexpected reasons
- Offsets → A common practice in many geodetic time series data
- Noise → white or colored noise



Trend analysis

The most basic form of time-series analysis examines trends that are sustained movements in the variable of interest in a specific direction.

- Trend analysis often takes one of two forms:
 1. Linear trend analysis, in which the dependent variable changes at a constant rate over time.

$$y_t = b_0 + b_1 t + \varepsilon_t$$

- Ex: if $b_0=3$ and $b_1=2.3$, then the predicted value of y after three periods is

$$y_t = 3 + 2.3(3) = 9.9$$

2. Log-linear trend analysis, in which the dependent variable changes at an exponential rate over time or constant growth at a particular rate

$$\ln(y_t) = b_0 + b_1 t + \varepsilon_t$$

- Ex: if $b_0=2.8$ and $b_1=1.4$, then the predicted value of y after three periods is

$$\ln(y_t) = 2.8 + 1.4(3) \xrightarrow{\text{yields}} y_t = 1096.63$$



Stationary Time Series

A stationary process is a stochastic process whose joint probability distribution does not change when shifted in time. Consequently, parameters such as the mean and variance, also do not change over time and do not follow any trends.

- The covariance is independent of t for each h

$$\text{COV}(x_t, x_{t-h}) \equiv E[(x_t - \mu)(x_{t-h} - \mu)]$$

- If h=0

$$\text{VAR}(x_t) \equiv E[(x_t - \mu)(x_t - \mu)]$$

- The mean is independent of t

$$E(x_t) = \mu$$



Covariance-stationary series

A time series is said to be covariance stationary if its mean and variance do not change over time.

- Time series that are not covariance stationary have linear regression estimates that are invalid and have no interpretation
- For a time series to be stationary
 1. The expected value of the series must be finite and constant across time.
 2. The variance of the series must be finite and constant across time.
 3. The covariance of the time series with itself must be finite and constant for all intervals over all periods across time.
- Visually, we can inspect the time-series model for a mean and variance that appear stationary as an initial screen for likely stationarity.



Why Stationary Time Series?

It is desire to have stationary time series

- Stationary time series have the best linear predictor.
- Nonstationary time series models are usually more difficult for prediction



Ergodic Time Series

Values of the process are measured over a finite period in time during one run of the experiment

- A stationary random process is also ergodic if every particular member of the ensemble is representative for all members. Thereby we are allowed, when estimating a certain parameter of the density function, to replace the ensemble average by the corresponding time average, obtained from the only one realization.
- Stationarity is thus a prerequisite for a process to be ergodic.
- Ergodicity is defined both for mean and (co)variances
- In general ergodicity of a random process can not be verified; it has to be assumed.



Autocovariance and Autocorrelation

The formal autocovariance between $y(t)$ and $y(t+\tau)$ for a process $y(t)$ is defined as

$$Q_{y_t y_{t+\tau}} = C\{\underline{y}(t), \underline{y}(t + \tau)\} = E\{(\underline{y}(t) - \mu)(\underline{y}(t + \tau) - \mu)\}$$

For a stationary random process the autocovariance function depends on the time difference between instant t and $t+\tau$, i.e.

$$Q_{y_t y_{t+\tau}} = Q_{yy}(\tau)$$

The autocorrelation between $y(t)$ and $y(t+\tau)$ for a process $y(t)$ is defined as

$$\rho(\tau) = \frac{Q_{yy}(\tau)}{Q_{yy}(0)}$$

where $Q_{yy}(0) = \sigma^2$ is the variance of the noise process



Covariance matrix based on autocovariance

Values of the process are measured over a finite period in time during one run of the experiment

$$Q = \begin{pmatrix} Q(t_1 - t_1) & Q(t_1 - t_2) & \cdots & Q(t_1 - t_m) \\ Q(t_2 - t_1) & Q(t_2 - t_2) & \cdots & Q(t_2 - t_m) \\ \vdots & \vdots & \ddots & \vdots \\ Q(t_m - t_1) & Q(t_m - t_2) & \cdots & Q(t_m - t_m) \end{pmatrix}$$

For equidistant sampling according to $t_i = t_o + iT$ we have

$$Q = \begin{pmatrix} Q(0) & Q(T) & \cdots & Q((m-1)T) \\ Q(T) & Q(0) & \cdots & Q((m-2)T) \\ \vdots & \vdots & \ddots & \vdots \\ Q((m-1)T) & Q((m-2)T) & \cdots & Q(0) \end{pmatrix}$$



Covariance matrix based on autocovariance

Structure of Covariance Matrix

$$Q_y = \begin{bmatrix} \sigma^2 & \sigma_1 & \sigma_2 & \cdots & \sigma_{m-1} \\ \sigma_1 & \sigma^2 & \sigma_1 & \ddots & \vdots \\ \sigma_2 & \sigma_1 & \sigma^2 & \ddots & \sigma_2 \\ \vdots & \ddots & \ddots & \ddots & \sigma_1 \\ \sigma_{m-1} & \cdots & \sigma_2 & \sigma_1 & \sigma^2 \end{bmatrix}$$

m number of (co) variance factors (Toeplitz matrix)



Basic noise processes

- In this section we discuss several different types of stochastic processes that are useful in modeling time series:
 - (1) A purely random process,
 - (2) An autoregressive process: $AR(1)$,
 - (3) A random constant process,
 - (4) A random walk noise process,



Different noise process

Purely random noise process

- A purely random process (white noise process) yields a sequence of uncorrelated (but not necessarily independent) random variables. For a zero mean noise process we have

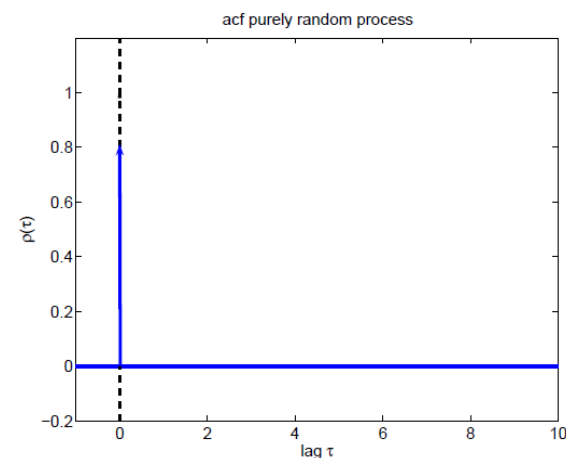
$$\underline{y}_t = \underline{e}_t$$

$$Q(\tau) = \begin{cases} \sigma^2 & \tau = 0 \\ 0 & \tau \neq 0 \end{cases}$$

$$\rho(\tau) = \begin{cases} 1 & \tau = 0 \\ 0 & \tau \neq 0 \end{cases}$$

- Autocovariance

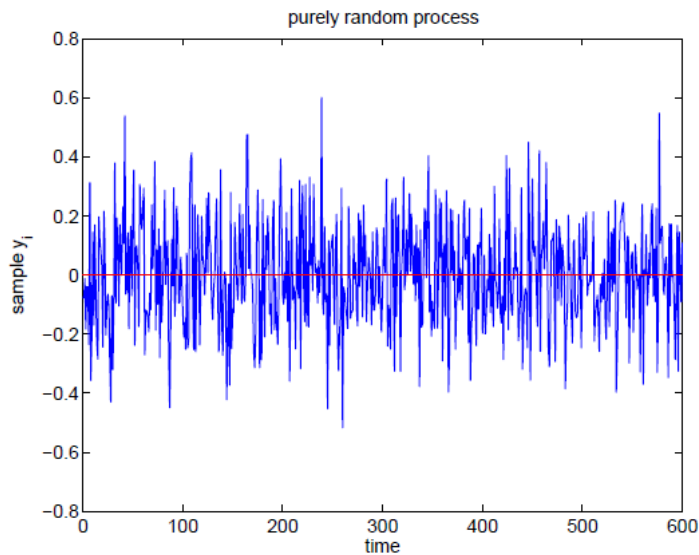
- Autocorrelation



Purely random noise process

Covariance matrix

$$E\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad D\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \sigma^2 \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}$$



Simulated data



Autoregressive noise process of order 1

- A zero mean first order auto-regressive noise process, abbreviated to AR(1), and also called a linear Markov process (if it is Gaussian), is given by the first order stochastic difference equation

$$\underline{y}_t - \beta \underline{y}_{t-1} = \underline{e}_t \quad \text{where} \quad 0 < \beta < 1$$

- Corresponding stochastic differential equation

$$\dot{\underline{y}}(t) + \alpha \underline{y}(t) = \underline{e}(t)$$

- Integration over interval t-1 to t, yields: $\underline{y}_t = \underbrace{e^{-\alpha \tau}}_{\beta} \underline{y}_{t-1} + \underbrace{\int_{t-1}^t e^{-\alpha(t-\nu)} \underline{e}(\nu) d\nu}_{\underline{e}_t}$
- Meaning of autoregressive



Autoregressive noise process of order 1

$$Q(\tau) = \sigma^2 \beta^{|\tau|} \quad \text{Autocovariance}$$

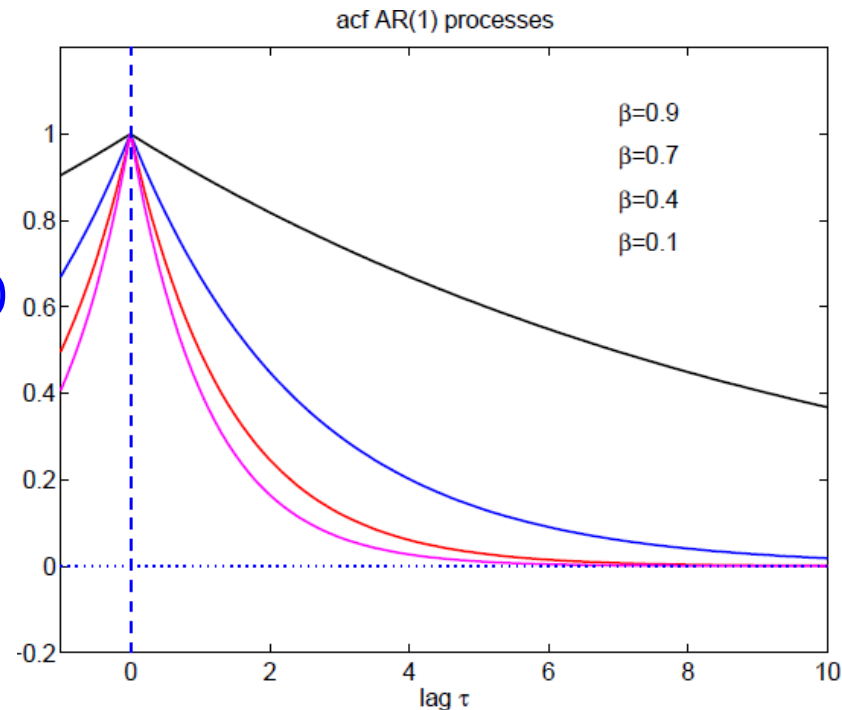
$\beta = e^{-\alpha}$, with $\alpha > 0$ and $\frac{1}{\alpha}$ correlation length (time)

$$\rho(\tau) = \beta^{|\tau|} \quad \text{Autocorrelation}$$

- Autocorrelation function of an autoregressive process of order 1: AR(1)

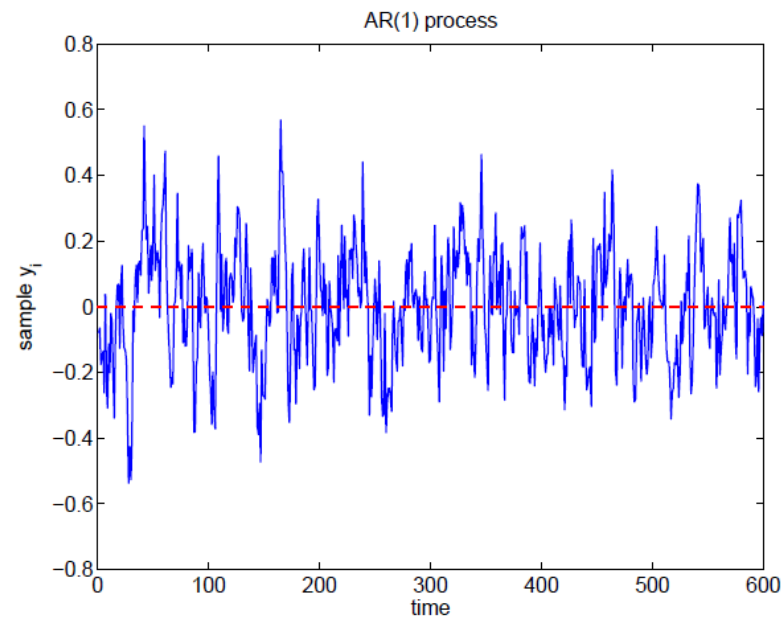
$$\beta = 0.1, 0.4, 0.7 \text{ and } 0.9$$

$$\frac{1}{\alpha} = 0.4, 1.1, 2.8 \text{ and } 9.5$$



Autoregressive noise process of order 1

- Time series of an AR(1) noise process, generated with first sample $y_1 \sim N(0, 0.18^2)$
- Remaining $(m-1)$ samples $e_i \sim N(0, 0.18^2(1-\beta^2))$ with $i = 2, \dots, m$
 $m = 600, \beta = 0.7$
- Time series is stationary
- Time correlation is visually seen in data
- This could be further verified by EACF



Autoregressive noise process of order 1

- For a zero mean first order autoregressive noise process $y(t)$, with $y_1, \dots, y_m$ equidistant ($T = 1$), the model of observation equations reads

$$E\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad D\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \sigma^2 \begin{pmatrix} 1 & \beta & \dots & \beta^{m-1} \\ \beta & 1 & \dots & \beta^{m-2} \\ \vdots & \vdots & \ddots & \vdots \\ \beta^{m-1} & \beta^{m-2} & \dots & 1 \end{pmatrix}$$

- The covariance matrix is a symmetric positive definite matrix (Toeplitz)
- By multiple application of

$$\underline{y}_t - \beta \underline{y}_{t-1} = \underline{e}_t$$

one obtains



Autoregressive noise process of order 1

$$\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix} = \begin{pmatrix} 1 & & & \\ \beta & 1 & & \\ \vdots & \vdots & \ddots & \\ \beta^{m-1} & \beta^{m-2} & \dots & 1 \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix}$$

which is inverted to

$$\begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix} = \begin{pmatrix} 1 & & & \\ -\beta & 1 & & \\ & \ddots & \ddots & \\ & & -\beta & 1 \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}$$

- Equivalent formulation

- Correlated/Uncorrelated

- Two extremes $\beta = 0$ or 1

- Application of error propagation law yields

$$D\left\{ \begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix} \right\} = \sigma^2 \begin{pmatrix} 1 & & & \\ & (1 - \beta^2) & & \\ & & \ddots & \\ & & & (1 - \beta^2) \end{pmatrix}$$

- Number of independent observations is less than m



Random constant noise process

- With $\beta = 1$, a zero mean first order autoregressive noise process reduces to either a random walk or a random constant noise process
- In this case the observation equations are

$$E\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad D\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \sigma^2 \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}$$

- The covariance matrix is a positive semi-definite matrix (rank equal to 1)
- Correlation length $\frac{1}{\alpha}$ is infinite: $\beta=e^{-\alpha} \rightarrow 1$ gives $\alpha \rightarrow 0$
- Random constant look like a systematic errors in all observations



Random constant noise process

- If we take $\beta = 1$, the previous relation become:

$$\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix} = \begin{pmatrix} 1 & & & \\ 1 & 1 & & \\ \vdots & \vdots & \ddots & \\ 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix}$$

which is inverted to

$$\begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix} = \begin{pmatrix} 1 & & & \\ -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}$$

- Application of error propagation law yields

$$D\left\{ \begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix} \right\} = \sigma^2 \begin{pmatrix} 1 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{pmatrix}$$

- Number of independent observations is only 1 (y_1)



Random constant noise process

- Stochastic difference equation becomes

$$\underline{y}_t - \beta \underline{y}_{t-1} = \underline{e}_t \rightarrow \underline{y}_t - \underline{y}_{t-1} = 0$$

- The stochastic differential equation (in continuous time) is

$$\dot{\underline{y}}(t) = 0$$

- Autocovariance and autocorrelation are

$$Q(\tau) = \sigma^2 \forall \tau$$

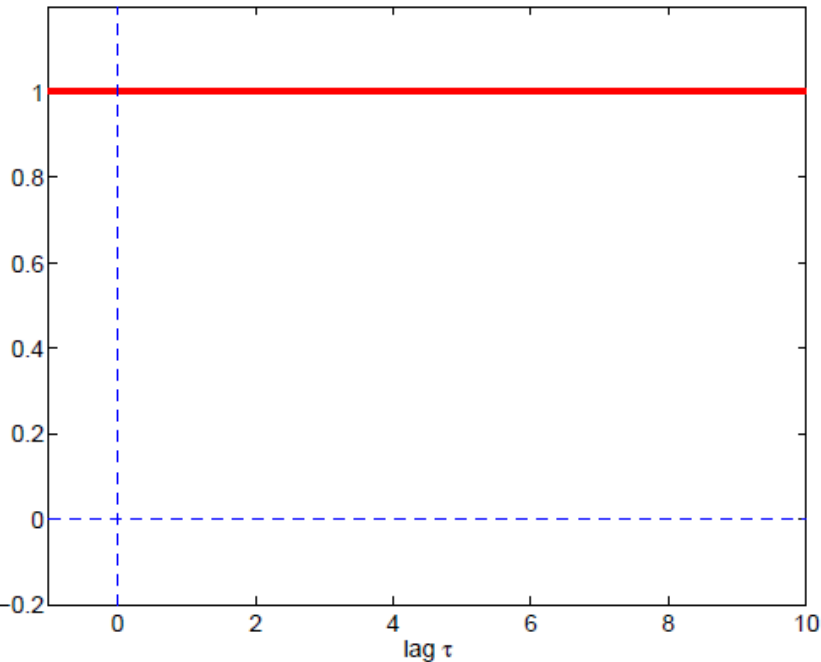
$$\rho(\tau) = 1 \forall \tau$$

- The correlation coefficient between any two samples equals $\rho = 1$



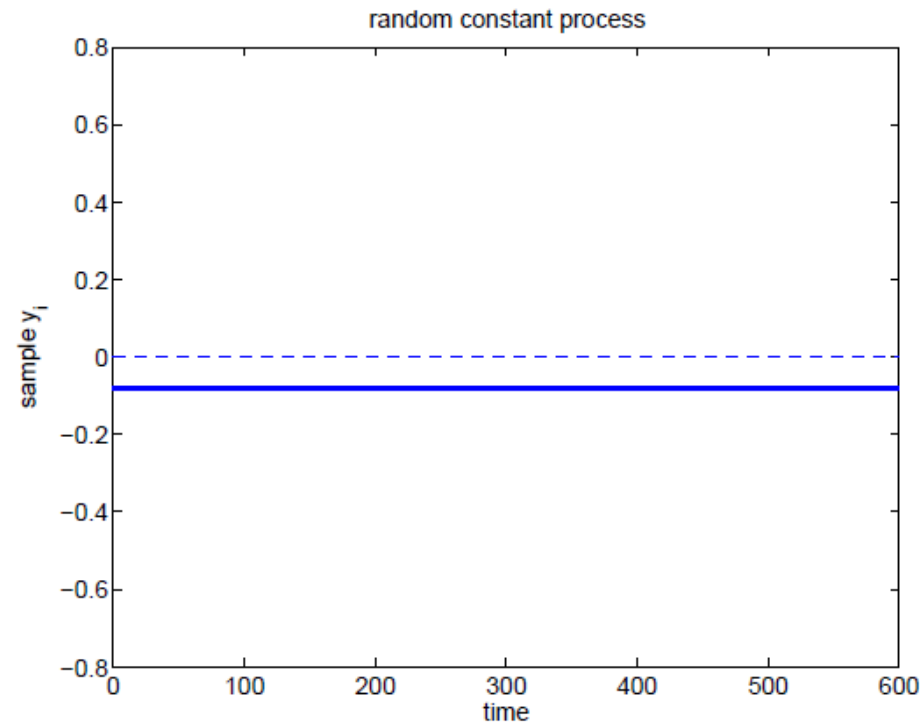
Random constant noise process

acf random constant process



- Autocorrelation function of a random constant process

- Time series of (only one realization of) a random constant noise process



Random walk noise process

- Substitution of $\beta = 1$ in the first order stochastic difference equation of a zero mean first-order autoregressive noise process, yields a zero mean random walk noise process

$$\underline{y}_t - \underline{y}_{t-1} = \underline{e}_t \longleftrightarrow \underline{e}_t \text{ is a stationary purely random noise process}$$

- The corresponding stochastic differential equation is

$$\dot{\underline{y}}(t) = \underline{e}(t)$$

- Model of observations of a random walk noise process is

$$E\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad D\left\{\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}\right\} = \sigma^2 \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 2 & \cdots & 2 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & \cdots & m \end{pmatrix}$$

- Random walk is not a stationary noise process



Random walk noise process

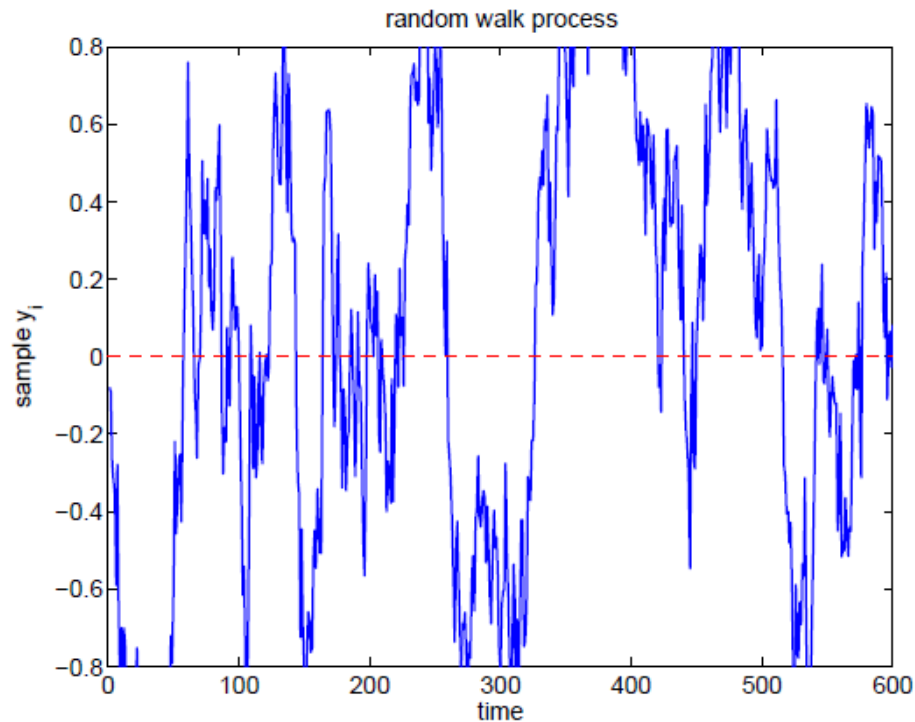
$$\begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix} = \begin{pmatrix} 1 & & & \\ 1 & 1 & & \\ \vdots & \vdots & \ddots & \\ 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix} \longleftrightarrow \begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix} = \begin{pmatrix} 1 & & & \\ -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \end{pmatrix} \begin{pmatrix} \underline{y}_1 \\ \underline{y}_2 \\ \vdots \\ \underline{y}_m \end{pmatrix}$$

- **We note** $\underline{y}_m = \underline{y}_1 + \sum_{i=2}^m \underline{e}_i$
- **Application of error propagation law yields**

$$D\left\{ \begin{pmatrix} \underline{y}_1 \\ \underline{e}_2 \\ \vdots \\ \underline{e}_m \end{pmatrix} \right\} = \sigma^2 \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}$$



Random walk noise process



- Time series of (one realization of) a random walk noise process

$\underline{e}_i \sim N(0, \sigma^2)$, with $i = 1, \dots, m$, $m = 600$ and $\sigma = 0.18$.



Other noise processes

- We may discuss other types of stochastic processes that are useful in modeling time series:
 - (1) a moving average (MA) noise process,
 - (2) an autoregressive process of order p : AR(p),
 - (3) an autoregressive moving average (ARMA) process
 - (4) an autoregressive integrated moving average (ARIMA) process
 - (5) A Gaussian noise process
 - (6) a power-law noise process
 - (7) Hole-effect noise process
 - (8) Spherical noise process
 - (9) Baarda & Alberta noise process



Other noise processes

- Autocorrelation functions of some noise processes:

Noise model	Cofactor matrix C_y
Power-law	$(\Delta t)^{-\frac{\alpha}{2}} T T^T, T = \begin{bmatrix} \Psi_0 & 0 & 0 & \dots & 0 \\ \Psi_1 & \Psi_0 & 0 & \dots & 0 \\ \Psi_2 & \Psi_1 & \Psi_0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Psi_n & \Psi_{n-1} & \Psi_{n-2} & \dots & \Psi_0 \end{bmatrix}, \Psi_n = \frac{\Gamma(n - \frac{\alpha}{2})}{n! \Gamma(-\frac{\alpha}{2})}$
First-order Gauss-Markov	$\exp(-\alpha \tau)$
Baarda & Alberda	$1 - \alpha \tau$
Hole-effect	$(1 - \alpha \tau) \exp(-\alpha \tau)$
Gaussian	$\exp(-\alpha \tau^2)$
Spherical	$1 - \frac{3}{2} \alpha \tau + \frac{1}{2} \alpha^3 \tau^3$



Time Correlation Estimation: Empirical ACF

Structure of Covariance Matrix:

$$Q_y = \begin{bmatrix} \sigma^2 & \sigma_1 & \sigma_2 & \cdots & \sigma_{m-1} \\ \sigma_1 & \sigma^2 & \sigma_1 & \ddots & \vdots \\ \sigma_2 & \sigma_1 & \sigma^2 & \ddots & \sigma_2 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \sigma_{m-1} & \cdots & \sigma_2 & \sigma_1 & \sigma^2 \end{bmatrix}$$

1. Side Diagonal
2. One variance factor
3. $m-1$ covariance factors
 $\sigma_{ij} = \sigma_{|j-i|} = \sigma_\tau$
4. Functional model:

m : total number of (co) variance factors



Time Correlation Estimation: Empirical ACF

Least-Squares Estimators:

$$\hat{\underline{\sigma}}_{\tau} = \frac{\sum_{i=1}^{m-\tau} \hat{\underline{e}}_i \hat{\underline{e}}_{i+\tau}}{m-\tau} \quad \tau = 0, 1, \dots, m-1 \quad \text{with } \hat{\underline{e}} = P_A^{\perp} \underline{y}$$

If $\tau = 0$ variance:

1. Unbiased Estimator
2. Precision of Estimator

Correlation coefficients (Auto-Correlation Function)

$$\hat{\underline{\rho}}_{\tau} = \frac{\hat{\underline{\sigma}}_{\tau}}{\hat{\underline{\sigma}}^2}, \quad \tau = 1, 2, \dots, m-1$$

$$\sigma_{\hat{\rho}_{\tau}}^2 = \frac{1}{m-\tau} + \frac{2\rho_{\tau}^2}{m}$$



Time Correlation Estimation: Empirical ACF

Empirical ACF

Auto-covariance function

$$\hat{Q}(\tau) = \frac{1}{m - |\tau|} \sum_{t=1}^{m-|\tau|} (y_t - \mu_1)(y_{t+|\tau|} - \mu_1) \quad \tau = 0, \pm 1, \pm 2, \dots, \pm(m-1)$$

← unbiased estimator

$$\hat{Q}'(\tau) = \frac{1}{m} \sum_{t=1}^{m-|\tau|} (y_t - \mu_1)(y_{t+|\tau|} - \mu_1) \quad \leftarrow \text{Biased estimator}$$

Auto-correlation function

$$\hat{\rho}(\tau) = \frac{\hat{Q}(\tau)}{\hat{Q}(0)} \quad \tau = 0, \pm 1, \pm 2, \dots, \pm(m-1)$$

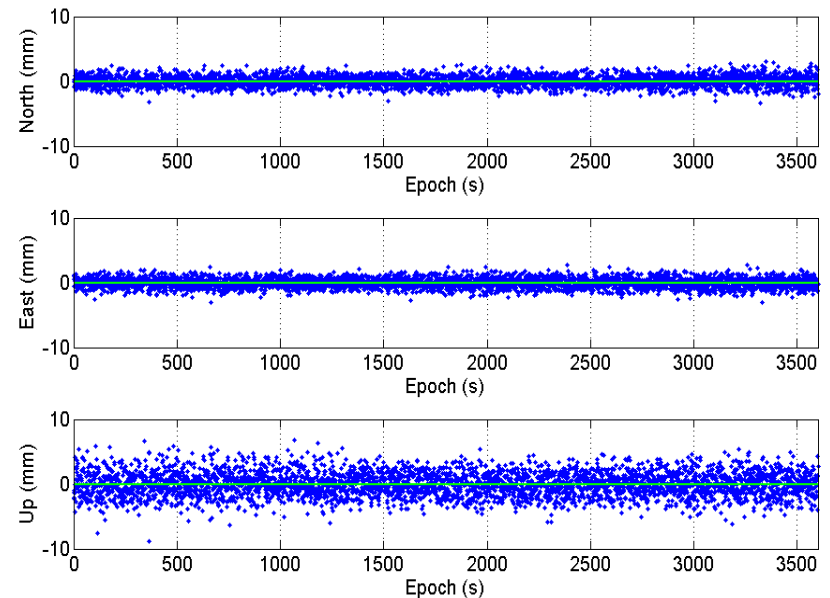
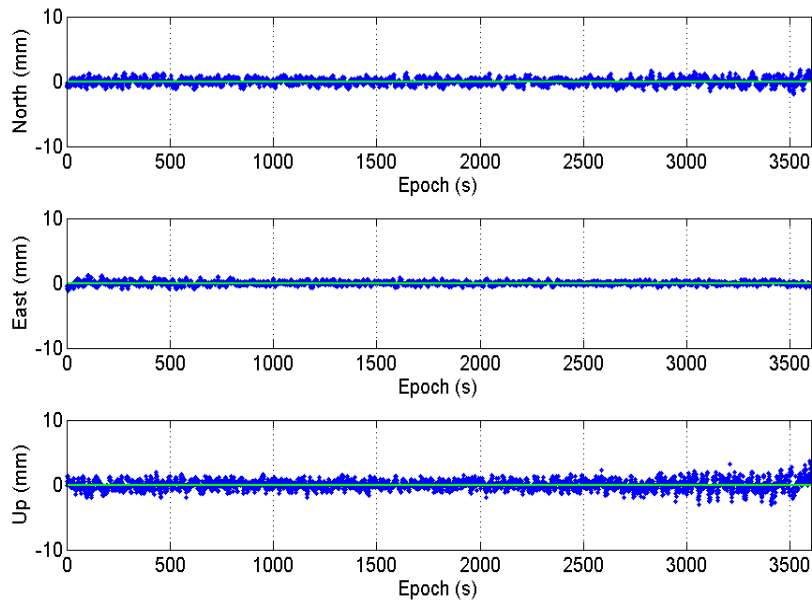
For AR(1) noise process we can estimate β

$$\hat{Q}(1) - \beta \hat{Q}(0) = 0 \Rightarrow \hat{\beta} = \frac{\hat{Q}(1)}{\hat{Q}(0)}$$



Position Time Series: GPS Data

Time series: GPS coordinates
Sampling Rate: 1 second
Time Span: 1 hour



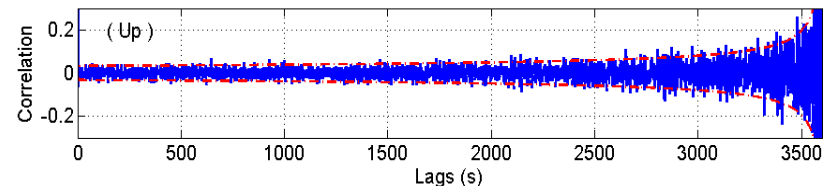
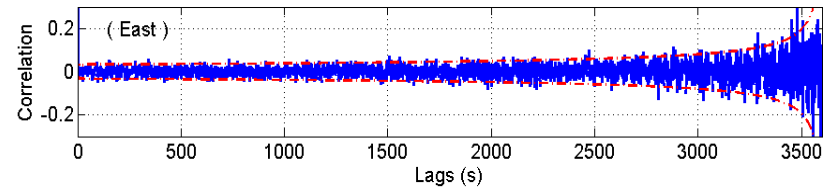
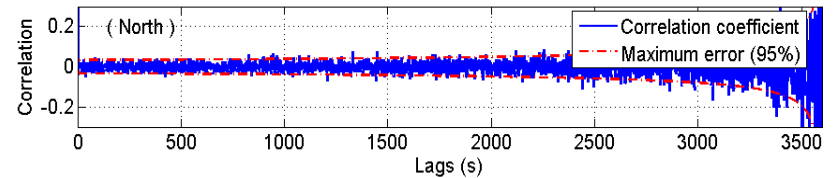
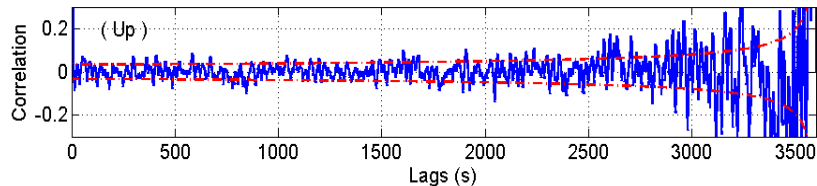
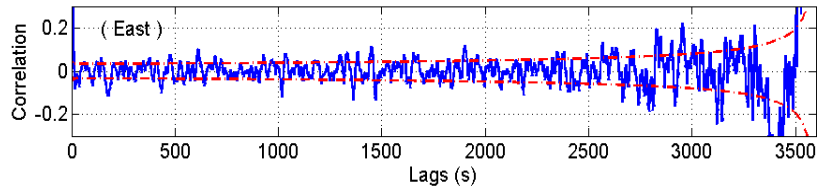
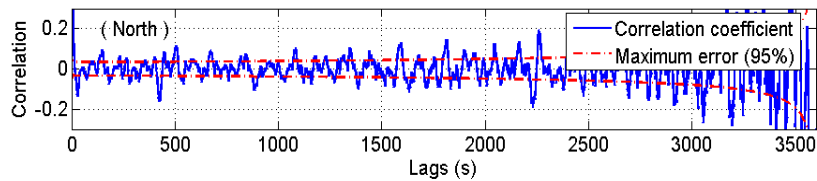
Position Time Series: GPS Data

Time series: **GPS coordinates**

Sampling Rate: **1 second**

Time Span: **1 hour**

(Empirical ACF)



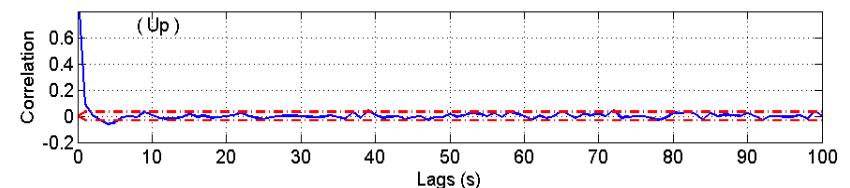
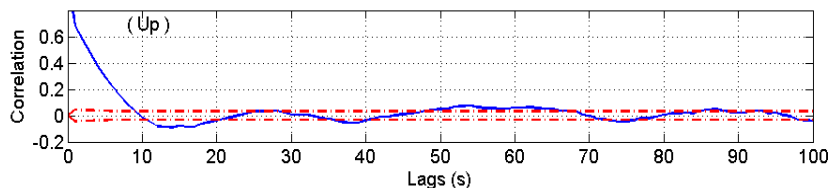
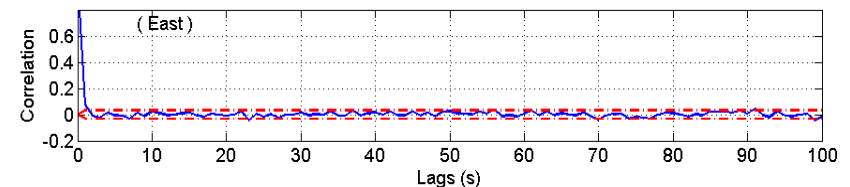
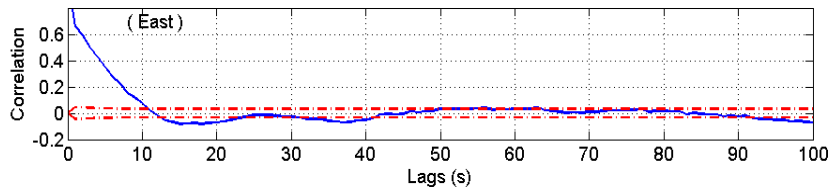
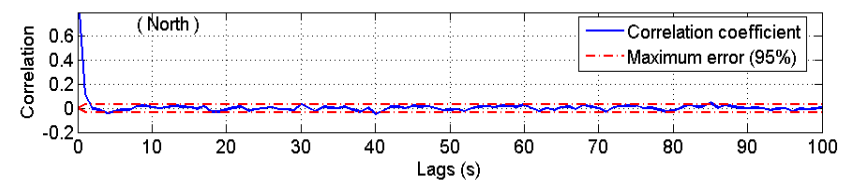
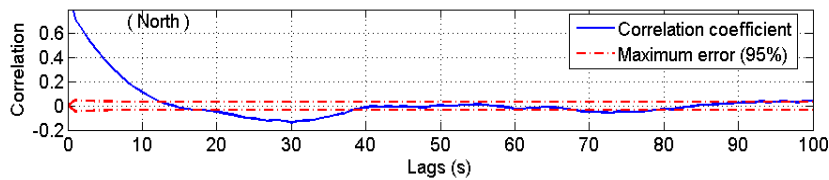
Position Time Series: GPS Data

Time series: **GPS coordinates**

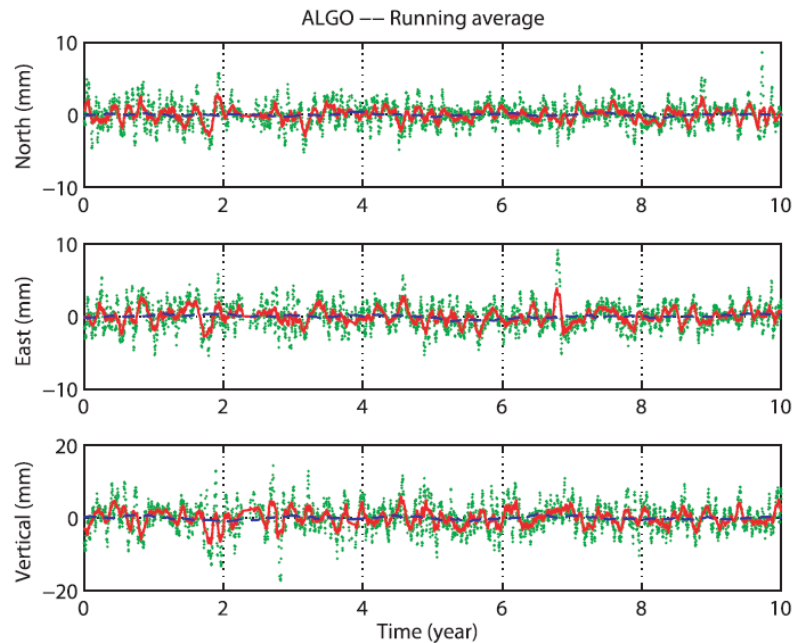
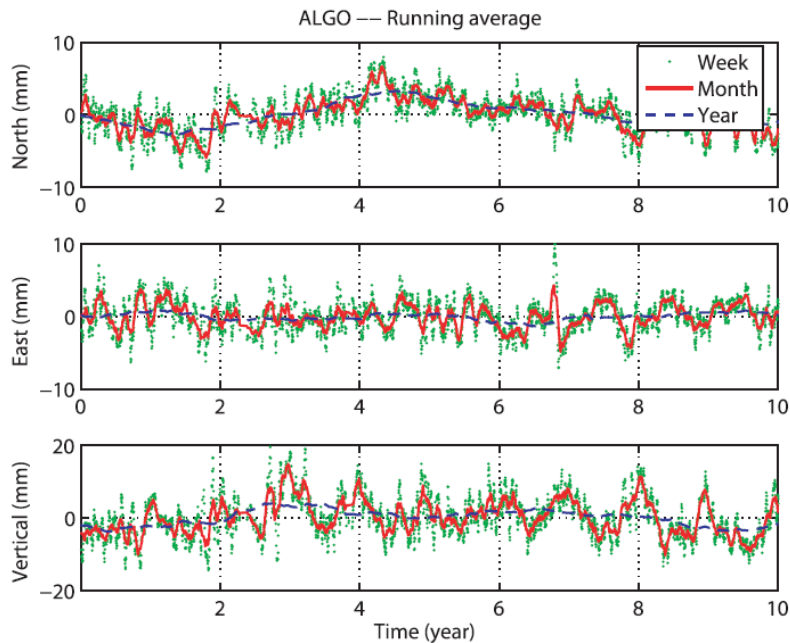
Sampling Rate: **1 second**

Time Span: **1 hour**

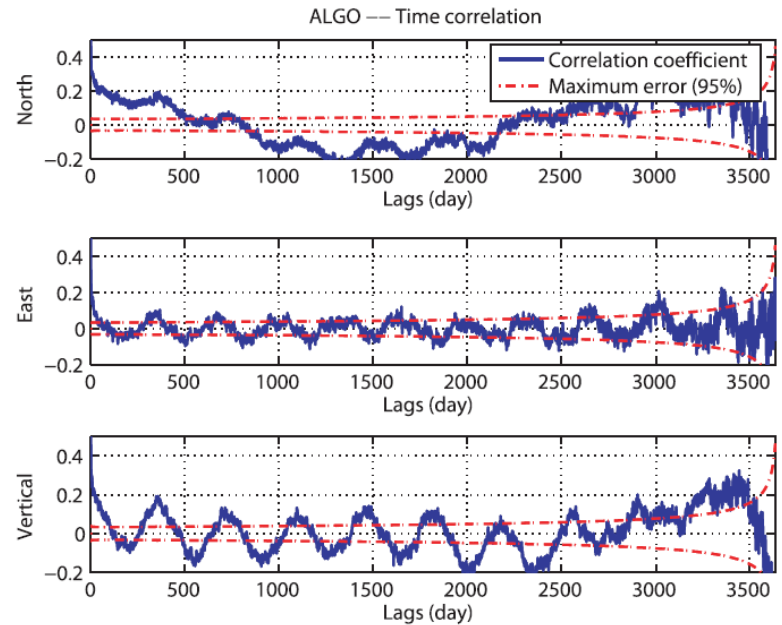
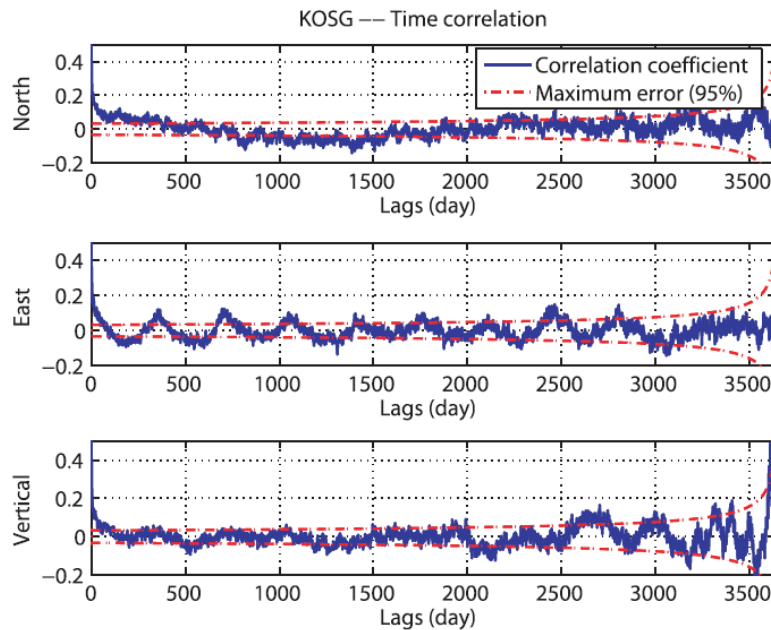
(Empirical ACF-zoom in)



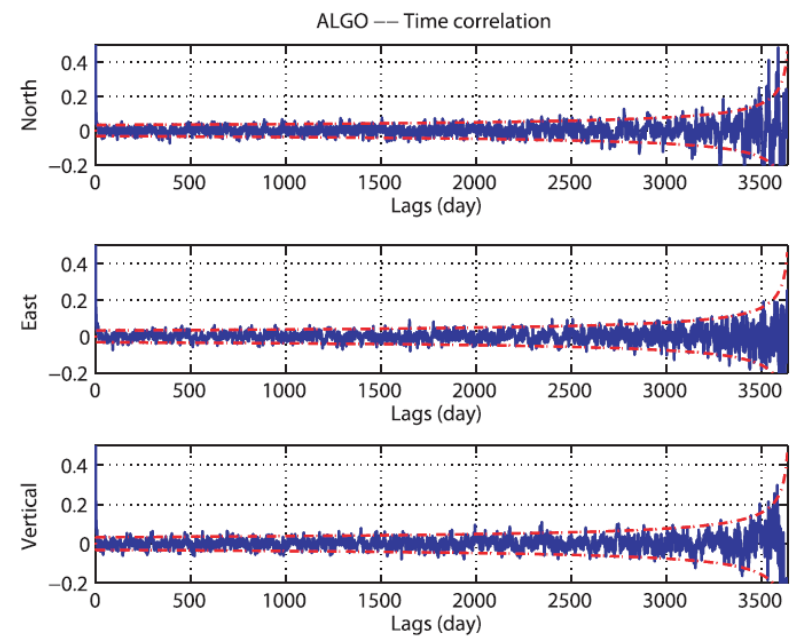
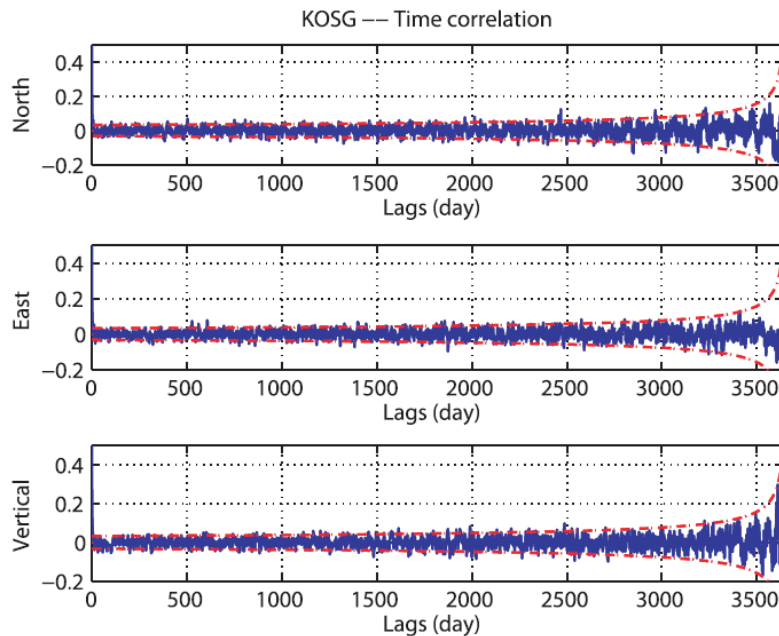
GPS coordinate time-series: Examples



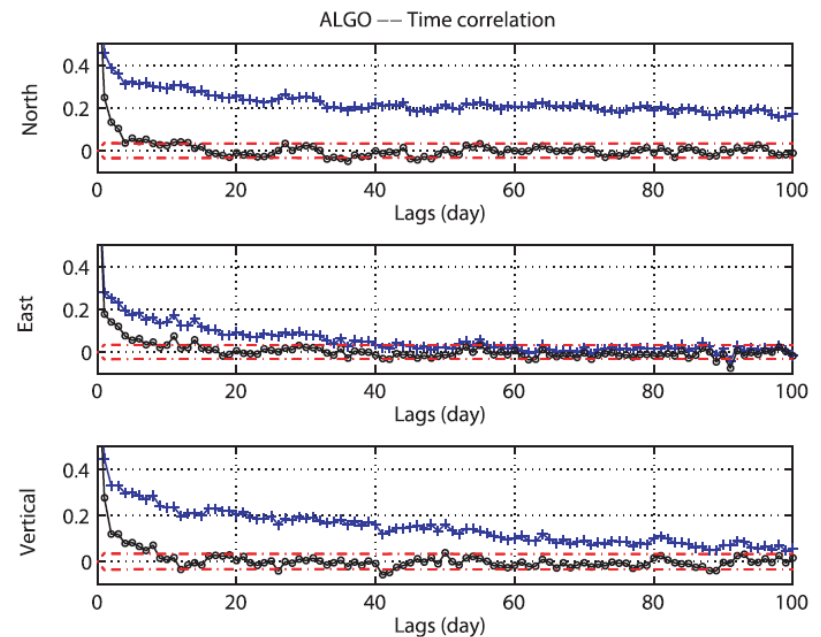
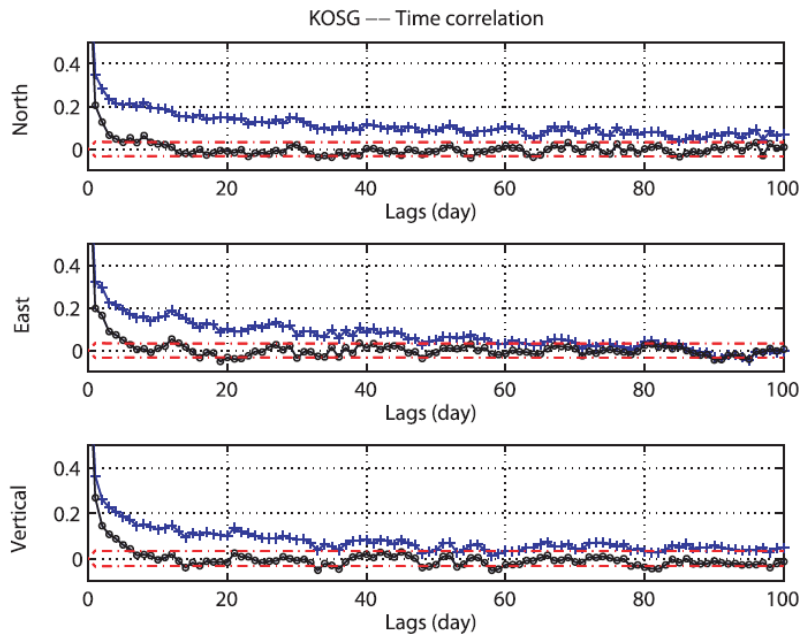
GPS coordinate time-series: ACF



GPS coordinate time-series: ACF (10 harmonics included)



ACF before and after 10 harmonics



Auto-correlation: 10 harmonics removed

